

MIND MAP — POSITIVE MATHEMATICAL COMMUNITIES



WHAT A POSITIVE MATHEMATICAL COMMUNITY LOOKS LIKE, SOUNDS LIKE, AND FEELS LIKE

Looks Like

A positive mathematical community is visually distinguished by the variety of strategies students use and the independence they display during problem solving. Dance and Kaplan (2018) in the titled section A Safe Space from Chapter One describe a first-grade classroom in which students approach a single math problem using tally marks, base ten blocks, ten frames, connecting cubes, equations, and written sentences, which is done all at the same time, and by student choice. This visible diversity of strategy use reflects a classroom where students have been trusted to select tools that make sense to them. Physically, the room looks collaborative: students work in partners and small groups, and a turn-and-talk structure means that pairs are seated face to face, knee to knee, as suggested by Dance & Kaplan (2018) in Chapter 2 from section Turn and Talks. According to Dance and Kaplan (2018), the teacher's role in this community is not to be the source of knowledge but rather a guide on the path toward knowledge construction, which is why Private think times or something along the lines should be included in this element, as suggested in Chapter 1 from section Constructing Knowledge.

Sounds Like

The sound of a positive mathematical community is one of productive student talk, respectful questioning, and collaborative sense-making. Dance and Kaplan (2018) illustrate this in Ms. Kaplan's classroom, where after a student named Harper shares her thinking, the teacher instructs turn to your neighbor and tell them what Harper's sentences mean in Chapter 1 from section Cooperative Learning and Mathematical Discourse. The guiding questions asked by the teacher define the sound of this community, which is not not here is the answer explanations, but rather questions such as if we have 30 here, what do we need now? This pushes students to reason for themselves as suggested by Dance & Kaplan (2018) in Chapter 1, from section Constructing Knowledge. Dance and Kaplan (2018) describe classrooms where students say, don't steal my thinking. This means calling out answers robs classmates of their chance to think, as suggested in Chapter 2, section Don't Steal My Thinking.

Feels Like

Perhaps the most critical dimension of a positive mathematical community is how it feels to students every day. Dance and Kaplan (2018) are explicit that students must feel safe to take risks, make mistakes, and grow — and that this sense of safety is the first major building block of a mathematical community, as suggested in Chapter 1 from section A Safe Space. Students feel that their ideas are valued and that classmates will celebrate rather than mock their mistakes. Dance and Kaplan (2018) frame respect as the foundation of this feeling, noting that when a respectful atmosphere exists, all learners feel that they and their ideas are important, which is highlighted in Chapter 2 and section Respect: Why It Matters. Dance and Kaplan (2018) warn that allowing students to give up or rushing in to save them does them a great disservice, and that classroom structures must actively encourage students to keep trying when things get tough, which is why it should feel like learners are empowered, as suggested in Chapter 1, from section Perseverance.

Connections Between Concepts

The safe space, which is Looks Like, directly enables perseverance, the Feels Like, which enables productivity because students only productively struggle when they trust it is safe to fail. Constructing knowledge, the Looks Like, is only possible through mathematical discourse, the Sounds Like. Thus, students build understanding by talking through ideas with classmates. Private think time, the Looks Like, protects every student's right to think, connecting directly to Feeling Respected and Valued, the Feels Like. Turn and talk, the Sounds Like, bridges independent thinking and cooperative learning.

REFERENCE

Dance, R., & Kaplan, T. (2018). *Thinking together: 9 beliefs for building a mathematical community*. Heinemann.